MIL-N-6748A

1 November 1968
SUPERSEDING
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MILITARY SPECIFICATION

NIPPLE; ELECTRICAL TERMINAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers one type of nipple for use as an insulator of electrical terminals.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

ZZ-R-765	Rubber, Silicone, Low-and -High-Temperature and Tear Resistant
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Box, Setup
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging

FSC 5975

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Federal (Continued)

PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Federal

FED-STD-601	Rubber; Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS25171	Nipple-Electrical Terminal

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications – The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Publications

D 149-61	Dielectric Strength of Electrical Insulating Material at Commercial Power Frequencies; test for

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction -The nipples furnished under this specification shall be a product which has been tested and passed the preproduction tests specified herein.

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3.2 Materials - Materials shall conform to ZZ-R-765 as specified in MS25171, and applicable portions of the requirements set forth herein.

3.3 Physical properties - Physical properties of the material used in the fabrication of nipples shall be as specified in Table I. The values required after oven aging and after oil immersion shall be based on the actual values obtained during tests on the unaged specimens.

3.4 Design and construction - The nipples shall be designed and constructed so as to conform to MS25171.

3.5 Workmanship - The nipples shall be manufactured and processed in accordance with the requirements of this specification. They shall be free from blisters, wrinkles, holes, dents, scratches and other defects which will affect serviceablility or appearance.

TABLE I

PHYSICAL PROPERTIES

Unaged:	
Hardness, ±5, Shore-A-durometer.	50
Tensile strength, minimum, pounds per square inch (psig).	700
Elongation, minimum, percent.	200
After oven aging:	
Hardness change, durometer, maximum.	±10
Tensile-strength change, maximum, percent.	-20
Elongation change, maximum, percent.	-40
After oil immersion:	
Hardness change, durometer, maximum.	-15 ±5

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TABLE I (Continued)

Tensile-strength change, maximum, percent.	-25
Elongation change, maximum, percent.	-20
Volume change, maximum, percent.	±15
Compression set, maximum, percent.	35
Brittle point, minimum, °C (°F).	-62.2(-80)
After water immersion for 70 hours at 100°C (212°F):	
Volume change, maximum, percent.	+5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection and testing of the nipples shall be classified as follows:

(a) Preproduction inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).
(1) Inspection of product for delivery (see 4.4.1)
(2) Inspection of preparation for delivery (see 4.4.4)

4.3 Preproduction inspection - Preproduction inspection shall be performed after the award of the contract on nipples representative of the production lot. Preproduction inspection shall consist of all inspection procedures of this specification (see 4.5).

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4.3.1 Preproduction samples - Preproduction samples shall be as specified in 4.4.2 and 4.4.2.1.

4.3.2 Preproduction specimens - Preproduction specimens shall consist of four (4) six-inch squares 1/8 to 1/10 inch thick from each lot of 10,000 nipples or less. The specimens shall be of the same compound, cure and physical properties as the end item.

4.3.3 Tests - The preproduction inspection of the nipple shall consist of all the tests of this specification as described in 4.5 (Test methods).

4.4 Quality conformance inspection -

4.4.1 Inspection of product for delivery - Inspection of product for delivery shall consist of the following tests:

(a) Examination of product

(b) Hardness

(c) Tensile strength

(d) Elongation

4.4.2 Sampling for inspection and acceptance - Samples for inspection and acceptance shall be selected from each inspection lot in accordance with the provisions set forth in MIL-STD-105. The acceptance number shall be zero for major defects. The AQL for minor defects shall be 6.5 percent defective.

4.4.2.1 Classification of defects - Defects found during examination shall be classified as specified in Table II.

TABLE II

CLASSIFICATION OF DEFECTS

ITEM	DEFECT	MAJOR	MINOR
Material	Failure to meet requirements.	X	
Appearance	Not uniform, not homogeneous.	X	
	Contains dirt, foreign matter.		X
	Parts not smooth, free from flash.		X
	Any tears, rips, cuts or holes.	X	
Size Measurements	Dimensions not within tolerances.	X	

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TABLE II (Continued)

ITEM	DEFECT	MAJOR	MINOR
Marking	Insufficient, incorrect, illegible.		X
Faulty Workmanship	Which might impair the usage of the nipples.	X	
	Which would not impair the usage of the nipples.		X

4.4.3 Inspection lot - An inspection lot shall consist of nipples of the same type produced under the same manufacturing conditions submitted at the same time. The units of inspection shall be as follows:

(a) Not less than 10 nipples from each lot of 1500 or less.

(b) Not less than one nipple per 2500 for all lots over 10,000.

4.4.4 Inspection of preparation for delivery - Sample items and packs shall be selected and inspected in accordance with MIL-P-116 to verify conformance with requirements of section 5 of this specification.

4.5 Test methods - Unless otherwise specified, all tests shall be performed in accordance with FED-STD-601.

4.5.1 Examination of product - Sample nipples shall be inspected to determine compliance with all requirements not covered by tests.

4.5.2 Hardness - Hardness shall be determined with a Shore A-2 durometer. Specimens shall be piled up to obtain a thickness at least 1/4 inch. The durometer reading shall be taken immediately after firm contact has been established between pressure foot and specimen. The hardness shall be the average of 5 readings.

4.5.3 Tensile strength and elongation - Tensile strength and elongation tests shall be conducted in accordance with Method 4111 and Method 4121 of FED-STD-601, using Die III to obtain specimens. Results of tests shall meet the requirements of Table I.

4.5.4 Flexibility at low temperature - Flexibility at low temperature test shall be conducted in accordance with Method 5211 of FED-STD-601. If the nipples cannot be attached to the electrical terminal without warming or damage, they shall be rejected.

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4.5.5 Oil immersion - Change in volume in the oil immersion tests shall be conducted in accordance with Method 6211 of FED-STD-601, using Medium No. 3 of Method 6001 for 22 ±1/4 hours at 100 ±1° C (212 ±2° F). Hardness, tensile strength and elongation shall be determined after allowing specimens to cool to room temperature using above methods. The resulting change in volume shall be in accordance with Table I.

4.5.6 Accelerated aging - Accelerated aging shall be conducted in accordance with Method 7001 of FED-STD-601 for 70 hours at 100 ±1° C (212 ±2° F). Specimens shall be bent 180° flat and examined. Hardness, tensile strength and elongation shall be determined, after allowing the specimens to cool to room temperature, using the above methods. Results shall be in accordance with Table I.

4.5.7 Water immersion - The water immersion test shall be conducted in accordance with Method 6211 of FED-STD-601 for 70 hours at 25 ±2.2° C (77 ±4° F). Results shall be expressed as percent change in volume and shall be in accordance with Table I.

4.5.8 Dielectric strength - The dielectric strength test shall be conducted in accordance with ASTM D 149-61 short time test. The dielectric strength shall be calculated as the break down voltage per mil of thickness and expressed as the average of three separate tests.

4.5.9 Compression set - The compression set test shall be conducted in accordance with Method 3311 of FED-STD-601. Results shall be in accordance with Table I.

4.5.10 Brittle point - The brittle point test shall be conducted in accordance with Method 5311 of FED-STD-601. Results shall be in accordance with Table I.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A -

5.1.1.1 Cleaning - Nipples shall be cleaned in accordance with MIL-P-116, Process C-1.

5.1.1.2 Drying - Nipples shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application - None required.

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5.1.1.4 Unit packaging - Unless otherwise specified, each nipple shall be individually packaged in accordance with MIL-P-116, Method III, insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.5 Intermediate packaging - Nipples packaged as described 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum tare and cube, and shall contain multiples of five unit packages, not to exceed 100 packages or ten pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 100 units.

5.1.2 Level C - Nipples shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This package may conform to the supplier's commercial practice for retail distribution when it meets the requirements of this level.

5.2 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A - The packaged nipples shall be packed in fiberboard containers conforming to PPP-B-636, Class Weather Resistant, Style optional, special requirement. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing all seams, conforming to PPP-T-60, Class 1, or PPP-T-76. Banding (reinforcing requirements) shall be applied in accordance with the appendix to PPP-B-636, using non-metallic or tape banding only.

5.2.2 Level B - Unless otherwise specified, the packaged nipples shall be packed in fiberboard containers conforming to PPP-B-636, Class Domestic, Style optional, and special use requirement. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C - The packaged nipples shall be packed in a shipping container in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This pack shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when it conforms to the requirements of this level.

5.3 Marking - In addition to any special marking required by the contract or order, each unit package, supplementary, intermediate, and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General - Exterior containers shall be of minimum tare and cube consistant with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible.

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5.5 Inspection - Inspection of military packaging shall be in accordance with MIL-P-116 (see Section 4).

6. NOTES

6.1 Intended use - The nipples covered by this specification are intended for use in aircraft to insulate and protect electrical terminal connections.

6.2 Ordering data - Procurement documents shall specify the following:

(a) Title, number and date of this specification

(b) Quantity of nipples

(c) Levels of packaging and packing required (see Section 5)

(d) MS25171 - (with required dash numbers)

6.3 Provisions for preproduction testing - Manufacture of any additional nipples on a particular contract should be suspended until the samples submitted for the preproduction tests are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these nipples from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.3.1 It is understood that the nipples supplied under contract or order will be identical with the corresponding preproduction sample in design, construction, quality, material, workmanship, and method of manufacture. Evidence of unauthorized change may constitute cause for rejection.

Custodians:
Army - AV
Navy - AS
Air Force - 85
Review activities:
Army
Navy
Air Force
User activities:
Army - ME, EL, MI
Navy - CG, MC
Air Force

Preparing activity:
Navy - AS
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-N-6748A NIPPLE; ELECTRICAL TERMINAL

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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